

Jungseob Lee

NLP Researcher · Ph.D. Student, Korea University

1995.10.11 | omanma1928@korea.ac.kr | js-lee-ai.github.io

js-lee-AI | LinkedIn | Google Scholar

Summary

Integrated M.S./Ph.D. researcher at the Korea University NLP Lab (Advisor: Prof. Heuiseok Lim), specializing in large language models and natural language processing — with a focus on agentic LLMs, LLM efficiency, and efficient reasoning. First-author research has been published at **EMNLP 2026** (Main), **ACL 2025** (Oral), and **Findings of ACL 2024**. Four papers were accepted at **EMNLP 2026** (2 Main, 2 Findings), with co-authored work at **ICLR 2026** and a series of preprints under review. Recipient of the **Best Paper Award (Top 1)** at HCLT 2025. Served as the post-training Project Manager and Korea University representative for **VAETKI**, Korea's national Sovereign AI Foundation Model. Author of 30+ peer-reviewed papers across top-tier NLP venues and journals.

Education

Integrated M.S./Ph.D. in Computer Science — *2021 – Present*

Korea University, Seoul, Korea — **Expected: Feb. 2027**

- NLP Lab, Advisor: Prof. Heuiseok Lim.
- Research focus: large language models, natural language processing, agentic LLM, LLM efficiency, and efficient reasoning.

Research Interests

Large Language Models · Natural Language Processing · Agentic LLM · LLM Efficiency · Efficient Reasoning

Work Experience & Major Projects

VAETKI — Korean Sovereign AI Foundation Model — *Jun 2025 – Jan 2026*

Post-training Project Manager — Korea University Representative — NC AI (NCSOFT)-led consortium

- Served as post-training PM representing Korea University in Korea's national **Sovereign AI Foundation Model** program, organized by the Ministry of Science and ICT — an approximately KRW 200 billion national initiative with staged six-month evaluations running through 2027.
- The Korea University team (Prof. Heuiseok Lim) was selected as a core research partner in the NC AI (NCSOFT)-led grand consortium of 14 industry–academia–research institutions and 40 adopters (including POSCO DX, Lotte Innovate, KAIST, ETRI, and MBC).
- Led the full post-training pipeline — Supervised Fine-Tuning (SFT), Reinforcement Learning, and dataset curation — for the VAETKI model.
- **VAETKI model:** an industry-specialized sovereign foundation model, open-sourced on Hugging Face on Dec 30, 2025. A 100B+ parameter Mixture-of-Experts architecture (11B active parameters at inference) with a custom Multi-head Latent Attention (MLA) reducing memory usage by up to 83%.

- Achieved state-of-the-art results on Korean language and culture (CLIcK), advanced reasoning (KoBALT), professional knowledge (KMMLU-Pro), and instruction following (IFEval).
- [Model] [Technical Report]
- Press: [Korea University] [AI Times] [Newstomato]

Cross-lingual Language Transfer for Large Language Models *2024 – 2025*
Lead Researcher (*first author*) Published at ACL 2025 (Oral)

- Proposed a cross-lingual optimization method that efficiently transfers English-centric LLMs to new target languages while preserving source-language ability; accepted as an **Oral** presentation at ACL 2025.
- [PDF]

Efficient Reasoning for Large Reasoning Models *2025 – Present*
Lead Researcher (*first author*) Findings of EMNLP 2026; preprints under review

- Leading a first-author line of work on training-free adaptive thinking budgets (DART) and stable efficiency training via adaptive correct-only rewards (ACOER) for large reasoning models; DART was accepted to **Findings of EMNLP 2026**.
- Co-first author of **ThinkFuse**, a trajectory-aware test-time fusion method for small reasoning models, accepted to **Findings of EMNLP 2026**.
- [DART] [ACOER] [Code]

Awards & Honors

Outstanding Reviewer Award *2026*
The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), San Diego
Recognized by the ACL program committee for review quality and service to the conference. [Certificate]

Best Paper Award (Top 1) *2025*
Annual Conference on Human and Cognitive Language Technology (HCLT 2025)
For "*KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*"; **J. Lee** (first author), M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, H. Lim — Korea University; Soongsil University; Human-Inspired AI Research.

Minister of Science and ICT Award *2023*
ICT Challenge 2023, Ministry of Science and ICT (organized by IITP)
Team M. Kang, **J. Lee**, S. Lee, for "*KoCheckGPT*" — Korean-specialized detection of LLM-generated documents for enterprise security; selected from 79 teams (263 participants) across 28 universities. [Press]

Publications

Underlined = author. * denotes equal contribution. Full profile on Google Scholar.

2026

- "*The Hallucination Signal Is a Mean Shift: Why Simple Probes Suffice*"
J. Lee, J. Seo, H. Lim — **EMNLP 2026** (Main) [arXiv]
- "*Language Chain in Alignment: Cross-Lingual Ranking Preference Optimization*"
S. Lee, M. Kim, **J. Lee**, H. Lim — **EMNLP 2026** (Main)
- "*DART: Draft-Agreement Routing for Training-Free Adaptive Thinking Budgets in Hybrid Reasoning Models*"
J. Lee, S. Hong, S. Lee, J. Seo, J. Son, S. Eo, C. Park, H. Park, H. Moon, H. Lim — **Findings of EMNLP 2026** [arXiv] [Code]
- "*ThinkFuse: Trajectory-Aware Test-Time Fusion for Small Reasoning Models*"
M. Kang*, **J. Lee***, J. Seo, H. Lim — **Findings of EMNLP 2026** (*equal contribution)

- "*Improving Semantic Proximity in Information Retrieval through Cross-Lingual Alignment*"
  S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim — **ICLR 2026** (Poster) [PDF]
- "*Towards Enhancing Natural Language Inference for Identifying Erroneous Outputs in Data-to-Text Generation*"
  C. Ok, E.K. Lee, H. Ju, **J. Lee**, D. Oh — **Knowledge-Based Systems** [DOI]
- "*Skin-Deep: A Geometric Diagnostic for Alignment Fragility in Large Language Model Representations*"
  D.J. Lee*, **J. Lee***, S. Lee, S. Hong, S. Son, S. Eo, J. Seo, H. Lim — **Findings of AACL-IJCNLP 2026** (*equal contribution) [arXiv] [Code]
- "*Structured Clinical Memory for Safety-Consistent Diagnosis and Recommendation Generation in Long Medical Dialogues*"
  D.J. Lee, **J. Lee**, E. Park, T. Whang, H. Lim — **IEEE BIBM 2026** (Short Paper)

2026 Preprints

- "*Predicting and Repairing Merge Collapse in Large Language Models*"
  J. Lee, S. Lee, S. Eo, H. Moon, J. Seo, H. Lim [arXiv] [Code]
- "*Distilling Directional Verification*"
  J. Lee, S. Eo, S. Hong, S. Lee, C. Park, J. Seo, H. Lim [arXiv] [Code]
- "*CAST: Cost-Aware Speculative Trees from One-Pass Block Drafters*"
  J. Lee, S. Eo [arXiv] [Code]
- "*Refusal Localizes, the Damage Relocates: Safety Layers Under Few-Sample Fine-Tuning*"
  J. Lee*, D.J. Lee*, S. Eo, S. Hong, S. Lee, H. Lim (*equal contribution) [arXiv] [Code]
- "*Recovering Off-Policy Supervision for Speculative Decoding*"
  J. Lee, C. Park, S. Eo, H. Moon [arXiv] [Code]
- "*Risk-Controlled Selective LLM Answering by Pricing Label-Free Checks*"
  D.J. Lee*, **J. Lee***, C. Park, H. Moon, H. Lim (*equal contribution) [arXiv] [Code]
- "*Where Activation Sparsity and KV-Cache Sparsity Cross in LLM Decoding*"
  J. Lee, S. Lee, S. Hong, S. Eo, H. Lim [arXiv] [Code]
- "*Faster Block-Diffusion Serving with Distribution-Free Risk Guarantees*"
  J. Lee, D.J. Lee, C. Park, S. Eo, H. Lim [arXiv] [Code]
- "*Vision Is Not Overhead: One-Pass Block Drafting for Lossless Speculative Decoding in Vision-Language Models*"
  J. Lee, S. Hong, D.J. Lee, C. Park, J. Seo, S. Eo, H. Lim [arXiv] [Code]
- "*Answer-Conditioned Chains of Thought Degrade Verifiable-Reasoning Distillation in Large Language Models*"
  J. Lee, S. Lee, S. Son, D.J. Lee, S. Han, S. Eo, H. Lim [arXiv] [Code]
- "*To Isolate or to Score? Model-Adaptive Assessment for Cost-Efficient Multi-Agent RAG*"
  J. Lee, C. Park, H. Lim [arXiv] [Code]
- "*Beyond Penalizing Mistakes: Stabilizing Efficiency Training in Large Reasoning Models via Adaptive Correct-Only Rewards*"
  J. Lee*, S. Lee*, S. Hong, M. Kim, C. Park, H. Lim (*equal contribution) [arXiv] [Code]
- "*RenderRank: Learning to Rerank Text with Compressed Visual Tokens*"
  S. Hong, Y. Jang, **J. Lee**, H. Moon, H. Lim [arXiv]
- "*LAMAR: An Open Language-Aware Multilingual Alignment Reranker*"
  S. Hong, Y. Jang, **J. Lee**, S. Lee, H. Lim [arXiv]

2025

- "*Cross-lingual Optimization for Language Transfer in Large Language Models*"
  J. Lee, S. Hong, H. Moon, H.S. Lim — **ACL 2025** (Oral) [PDF]
- "*KULLM-RAG: A RAG-Specialized Large Language Model with Intelligent Self-Verification*"
  J. Lee, M. Kim, J. Yoon, S. Hong, Y. Jang, S. Lee, J. Seo, C. Park, J. Park, H. Lim — Annual Conference on Human and Cognitive Language Technology (HCLT 2025) — **Best Paper Award**
- "*An Analysis on Language Transfer of Pre-trained Language Model with Cross-lingual Post-training*"
  S. Son*, C. Park*, **J. Lee***, M. Shim, C. Lee, Y. Jang, J. Seo, J. Lim, H. Lim — **Expert Systems with Applications** (*equal contribution) [DOI]
- "*VAETKI Technical Report*"
  NC AI Consortium — Technical Report [PDF]

2024

- "*Length-aware Byte Pair Encoding for Mitigating Over-segmentation in Korean Machine Translation*"
  J. Lee, H. Moon, S. Lee, C. Park, S. Eo, H. Ko, J. Seo, S. Lee, H.S. Lim — **Findings of ACL 2024** [PDF]

- "*Analysis of the Effectiveness of Model, Data, and User-Centric Approaches for Chat Application: A Case Study of BlenderBot 2.0*"
  C. Park*, **J. Lee***, S. Son, K. Park, J. Jang, H. Lim — **Applied Sciences** (*equal contribution) [DOI]

2023

- "*PEEP-Talk: A Situational Dialogue-based Chatbot for English Education*"
  S. Lee, Y. Jang, C. Park, **J. Lee**, J. Seo, H. Moon, S. Eo, S. Lee, B. Yahya, *et al.* — **ACL 2023** (System Demonstrations) [PDF]
- "*A Survey on Evaluation Metrics for Machine Translation*"
  S. Lee*, **J. Lee***, H. Moon, C. Park, J. Seo, S. Eo, S. Koo, H. Lim — **Mathematics** (*equal contribution) [PDF]
- "*Doubts on the Reliability of Parallel Corpus Filtering*"
  H. Moon, C. Park, S. Koo, **J. Lee**, S. Lee, J. Seo, S. Eo, Y. Jang, H. Kim, H. Lee, *et al.* — **Expert Systems with Applications** [DOI]
- "*KoCheckGPT: Korean LLM Written Document Detector*"
  M. Kang, **J. Lee**, S. Lee, S. Hong, J. Park, H. Lim — Annual Conference on Human and Language Technology (HCLT 2023) [PDF]

2022

- "*QUAK: A Synthetic Quality Estimation Dataset for Korean-English Neural Machine Translation*"
  S. Eo, C. Park, H. Moon, J. Seo, G. Kim, **J. Lee**, H.S. Lim — **COLING 2022** [PDF]
- "*Empirical Analysis of Noising Scheme based Synthetic Data Generation for Automatic Post-editing*"
  H. Moon, C. Park, S. Lee, J. Seo, **J. Lee**, S. Eo, H.S. Lim — **LREC 2022** [PDF]
- "*K-NCT: Korean Neural Grammatical Error Correction Gold-standard Test Set Using Novel Error Type Classification Criteria*"
  S. Koo, C. Park, J. Seo, S. Lee, H. Moon, **J. Lee**, H. Lim — **IEEE Access** [PDF]
- "*Focus on FoCus: Is FoCus Focused on Context, Knowledge and Persona?*"
  S.Y. Lee*, **J. Lee***, C. Park, S. Eo, H. Moon, J. Seo, J. Park, H.S. Lim — 1st Workshop on Customized Chat Grounding Persona and Knowledge (*equal contribution) [PDF]
- "*There is No Rose Without a Thorn: Finding Weaknesses on BlenderBot 2.0*"
  J. Lee, M. Shim, S. Son, C. Park, Y. Kim, H. Lim — arXiv preprint [arXiv]
- "*Language Chameleon: Transformation Analysis between Languages Using Cross-lingual Post-training*"
  S. Son*, C. Park*, **J. Lee***, M. Shim, C. Lee, Y. Jang, J. Seo, H. Lim — arXiv preprint (*equal contribution) [arXiv]
- "*Korean and Multilingual Language Models Study for Cross-lingual Post-training* (*XPT*)"
  S. Son, C. Park, **J. Lee**, M. Shim, C. Lee, K. Park, H. Lim — **Journal of the Korea Convergence Society** [PDF]
- "*Policy-based Performance Comparison Study of Real-time Simultaneous Translation*"
  J. Lee, H. Moon, C. Park, J. Seo, S. Eo, S. Lee, S. Koo, H. Lim — **Journal of the Korea Convergence Society** [PDF]
- "*SaJuTeller: Conditional Generation Deep-Learning based Fortune Telling Model*"
  H. Moon, **J. Lee**, J. Seo, S. Eo, C. Park, W. Kim, J. Park, H. Lim — Annual Conference on Human and Language Technology (HCLT 2022) [PDF]
- "*Korean Language Model Construction and Comparative Analysis with Cross-lingual Post-Training* (*XPT*)"
  S. Son, C. Park, **J. Lee**, M. Shim, S. Lee, J.W. Lee, A. So, H. Lim — Annual Conference on Human and Language Technology (HCLT 2022) [PDF]
- "*DART: Data Augmentation using Retrieval Technique*"
  S. Lee, J. Seo, **J. Lee**, M. Kang, H. Moon, C. Park, D. Jung, J. Lee, K. Park, *et al.* — Annual Conference on Human and Language Technology (HCLT 2022) [PDF]

2021

- "*Empirical Study on BlenderBot 2.0's Errors Analysis in Terms of Model, Data and Dialogue*"
  J. Lee, S. Son, M. Shim, Y. Kim, C. Park, A. So, J. Park, H. Lim — **Journal of the Korea Convergence Society** [PDF]

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